

GANESHAMOORTHY ACKAASH | SOFTWARE ENGINEER INTERN

+94701705848 ackaash.20240491@iit.ac.lk linkedin.com/in/ackaash-ganeshamoorthy github.com/ackaashg-eng

ABOUT ME

Passionate and motivated Software Engineering undergraduate with hands-on experience developing scalable software solutions, RESTful APIs, and AI-powered applications by writing clean, testable and maintainable code following SOLID principles. Proficient in Java, Python, FastAPI, AWS, Docker, SQL, and modern software development practices, with a strong foundation in object oriented programming(OOP), data structures, and system design. Experienced in building end to end projects involving ML, NLP, cloud deployment, and database integration. A fast learner with following Agile and scrum practices, strong analytical and problem-solving abilities, eager to contribute to innovative engineering teams while continuously expanding technical expertise and delivering high-quality, maintainable software.

EDUCATION

BSc (Hons) in Computer Science	2025 - Present	G.C.E A/L Maths Stream	2022
University Of Westminster, Informatics Institute of Technology Colombo		Zahira College Matale	

TECHNICAL SKILLS

- Languages:** Python, JavaScript,TypeScript, SQL, Java
- Frontend & Design:** React JS, React Hooks, Context API, Tailwind CSS, Responsive Design, Figma, Next.js, HTML5, Bootstrap
- Backend & APIs:** Node.js, FastAPI, RESTful APIs, Express.js,
- Authentication:** JWT, OAuth
- Databases:** MySQL, PostgreSQL, MongoDB Atlas, JSON-based Storage
- BaaS / Cloud:** Supabase, Firebase, AWS
- DevOps:** Docker, GitHub Actions, CI/CD Pipelines, Microservices, IoT Integration
- Tools:** Git, GitHub, Postman, Linux, VS Code, IntelliJ, Cursor, Claude

Professional Experience | Projects

E-Commerce Management System – Full Stack Application 2026

Designed and developed a full-stack e-commerce application that enables customers to browse products, place orders through the website.

Key Technologies: Java, Spring Boot, Spring Security, React.js, MySQL, JWT, Docker, AWS EC2, GitHub Actions, Postman

- Built RESTful APIs using Java Spring Boot to manage products, categories, shopping carts, orders, and user accounts.
- Implemented secure user authentication and role based authorization using Spring Security and JWT, providing separate access for administrators, sellers, and customers.
- Developed a responsive React.js frontend that communicates with backend APIs, delivering a seamless user experience across product browsing, checkout, and order management.
- Built unit and integration tests using JUnit and Mockito for product, cart, and order service layers, achieving reliable and to maintain business logic
- Containerized the application using docker to deploy it in AWS EC2 server.

Hotel Booking System - Full Stack Application 2025

Built a production-grade fullstack application from concept to deployment.

Key Technologies: React, Node.js, Express.js, MongoDB, JWT, bcrypt, REST API, Docker, CI/CD

- Developed React frontend, Express backend, and MongoDB database.
- Implemented JWT authentication and bcrypt password hashing across endpoints.
- Validated APIs using Postman with GET, POST, PUT, DELETE methods.
- Built a Admin dashboard using React and Tailwind CSS to monitor rooms and bookings.
- Designed MongoDB collections and implemented database operations for users, hotels, and booking data, supporting efficient application workflows. Containerized for deployment using Docker.

Intelligent Document Assistant (RAG Chatbot) 2026 - PRESENT

Engineered a Retrieval Augmented Generation chatbot capable of answering questions from uploaded documents with contextual accuracy.

Key Technologies: Python, LangChain, FastAPI, OpenAI API, ChromaDB, React, Docker, Claude

Accomplishments:

- Developed a responsive React interface with reusable components for document upload, knowledge-base management, and chatbot interactions.
- Implemented a retrieval pipelines using LangChain and ChromaDB, performing document parsing, chunking, embedding generation, and vector similarity search for context-aware question answering.
- Utilized OpenAI embedding and LLM APIs to generate semantic document representations and produce accurate responses from retrieved content.
- Developed 5 REST API endpoints supporting authentication, document management, and AI inference, that enabled users to upload documents, manage knowledge bases, and interact with the chatbot.

AI-Powered Aphasia Therapy Robot- AI 2026 - Present

Engineered the backend software for an AI Aphasia robot, bridging a React dashboard and FastAPI backend with Supabase to execute real-time AI speech workflows. Designed and implemented a multi-agent architecture speech therapy system using LangGraph featuring four specialized agents.

Key Technologies: Python, PyTorch, Hugging Face, FastAPI, OpenAI API, AWS, Docker, React JS, Tailwind CSS, FastAPI, Supabase, PostgreSQL, CI/CD

- Integrated pre-trained Wav2Vec2 and G2P transformer models from Hugging Face to perform phoneme-level speech analysis and classification, producing structured feedback across three pronunciation error categories.
- Containerized and deployed scalable REST API services on AWS EC2 using FastAPI and Docker.
- Designed a multi-agent architecture with agents for audio analysis, phoneme detection, error correction, and feedback generation with real-time TTS output.
- Worked following Agile and Scrum framework, participating in sprint planning and daily standups to deliver features iteratively.
- Containerized and deployed scalable REST API services on AWS using FastAPI and Docker.

AWARDS & LEADERSHIP

- Runner-up - ADAPT Hackathon
- Cursor
- Ideathon - ThinkWave 3.0
- Hult Prize
- Code Sprint
- Toastmaster
- Speaker of Student Parliament
- Club Member - AI and Robotics Society (AIRS)

NON-TECHNICAL SKILLS

- Leadership
- Analytical critical thinking
- Time Management
- Quick and adaptive learner
- Team collaboration
- Strong communication
- Attention to detail

REFERENCES

G. Abirame

Senior Software Engineer
Synergen Health Pvt Ltd
abirameg@gmail.com

Mr. Saadh Jawwadh

Lecturer
IIT Colombo
saadh.j@iit.ac.lk